

Conditional Workflows in LangGraph | Agentic AI using LangGraph

1. Recap

- **Sequential workflow:** Tasks run one after another.
- **Parallel workflow:** Independent tasks run at the same time.
- **Conditional workflow:** The next step depends on a **condition or result**.

Basic idea:

Input → Decision → Path A / Path B

2. Conditional Workflow / QE

A **conditional workflow** uses a condition to decide which node should execute next.

Example:

Generate Answer → Quality Check

- If quality is **good** → Final Answer
- If quality is **poor** → Improve Answer

This is called **branching** because the workflow can take different paths.

Quality Evaluation (QE)

QE checks whether the generated result meets predefined requirements.

For example, check:

- Is the answer relevant?
- Is it complete?
- Is the quality acceptable?
- Does it follow the required format?

The result of the evaluation determines the next step.

3. LLM — Review Handling

An LLM can be used to **evaluate another LLM's output** and decide what should happen next.

Example:

Generate Response



LLM Reviewer



Is response acceptable?

- **Yes** → Final Response
- **No** → Review/Improve Response → Check Again

This creates an **adaptive workflow** where the path depends on the LLM's evaluation.

Key takeaway: Conditional workflows allow LangGraph to **make decisions during execution instead of following one fixed path.**